

# IRREDUCIBILITY OF THE POLYNOMIAL $x^{n-1} + 2x^{n-2} + \cdots + n$

## OVERVIEW

This text explains a proof that, for every integer  $n \geq 2$ , the polynomial

$$f(x) = x^{n-1} + 2x^{n-2} + \cdots + (n-1)x + n$$

is irreducible over  $\mathbb{Q}$ . Equivalently, its reversal  $1+2x+\cdots+nx^{n-1}$ , the derivative of  $1+x+\cdots+x^n$ , is irreducible. The text is meant to be read and checked rather than to be a research article: every step is a separate result with its own proof, and notation used in more than one place is collected at the end of the introduction.

The idea is to suppose that  $f = gh$  with  $g, h$  monic integer polynomials, and to look at one integer, the resultant  $\text{Res}(g, h)$ , in two different ways.

- Measured with the ordinary absolute value, it is *large*. The roots of  $f$  lie in a thin annulus just outside the unit circle. This controls the product of pairwise distances between the roots of  $g$  and gives an upper bound on  $|\text{Disc}(g)|$ ; an exact identity linking  $\text{Disc}(g)$  and  $\text{Res}(g, h)$  then forces  $|\text{Res}(g, h)|$  up (Section 4).
- Measured one prime at a time, it is *small*. For a prime  $p$ , the  $p$ -adic analysis describes disjoint packets of roots around roots of unity. A factor with rational coefficients takes whole packets, except that a quartic packet at  $p = 2$  may split into two quadratics. An elementary max-cut estimate, including this exception, bounds how much each prime can contribute to the resultant (Section 3).

The two bounds conflict as soon as both constant terms  $g(0)$  and  $h(0)$  are at least 5 (Section 5). The primes that matter are controlled by the integer  $J = g_1 - g(0)g_{\deg g-1}$ , built from three coefficients of  $g$ : it is small, yet divisible by every prime factor of  $h(0)(n+1)$ . The same integer rules out constant terms at most 4 (Section 6).

The argument uses  $n \geq 1000$  only through numerical estimates; for  $2 \leq n < 1000$  the theorem is checked by computer (Appendix A). Section 7 explains which additional arguments cover these  $n$  by hand. That section is not needed for the main argument and can be skipped.

The prerequisites are complex numbers, polynomials and elementary number theory. Section 3 also uses  $p$ -adic valuations; the facts needed are collected in Subsection 3.1.

## 1. INTRODUCTION

Fix an integer  $n \geq 2$  and put

$$(1.1) \quad f(x) = \sum_{k=1}^n kx^{n-k} = x^{n-1} + 2x^{n-2} + \cdots + (n-1)x + n.$$

Its reversal  $x^{n-1}f(1/x) = 1 + 2x + \cdots + nx^{n-1}$  is the derivative of  $1 + x + \cdots + x^n$ , so the irreducibility of the two polynomials is equivalent.

**Theorem 1.1.** *For every integer  $n \geq 2$  the polynomial  $f$  is irreducible in  $\mathbb{Q}[x]$ , and hence in  $\mathbb{Z}[x]$ .*

When  $p = n+1$  is prime,  $f(1+y) = \sum_{k=0}^{n-1} \binom{p}{k+2} y^k$  is  $p$ -Eisenstein, and the theorem is classical in that case.

**Small  $n$ .** For  $2 \leq n < 1000$  the theorem was verified by computer: the polynomial  $f$  was factored over  $\mathbb{Z}$  with FLINT (via `python-flint` 0.9.0) for each such  $n$ , and in every case it is irreducible (Appendix A). *From now on we assume  $n \geq 1000$ .* This assumption is used only through a few numerical estimates, and it removes several case distinctions.

**Strategy.** Write  $F(x) = (x-1)^2 f(x) = x^{n+1} - (n+1)x + n$ . Suppose  $f = gh$  with  $g, h \in \mathbb{Z}[x]$  monic and nonconstant, and put  $a = g(0)$ ,  $b = h(0)$ ,  $d = \deg g$ . Then  $ab = n$  and  $\gcd(a, b) = 1$ .

- (1) All roots of  $f$  lie in  $1 < |z| < (2n+1)^{1/n}$  (Lemma 2.1).
- (2) The derivative  $f'$  takes the simple value  $n(n+1)/(\alpha(\alpha-1))$  at every root  $\alpha$ , which gives the exact identity  $|\text{Disc}(g)| |\text{Res}(g, h)| g(0) g(1) = (n(n+1))^d$  (Lemma 2.5).
- (3) Since the roots of  $g$  are confined to a thin annulus,  $|\text{Disc}(g)|$  is small (Lemma 4.1); hence  $|\text{Res}(g, h)|$  is large.
- (4) For each prime  $p \mid n(n+1)$ , the unit roots of  $f$  in  $\overline{\mathbb{Q}}_p$  form packets around the  $m$ -th roots of unity ( $p \nmid m$ ). A rational factor takes whole packets, except that a quartic packet at  $p = 2$  may split into two quadratics. A max-cut argument then gives  $v_p(\text{Res}(g, h)) \lesssim (n+1)/p$ , and  $v_p(\text{Res}(g, h)) = 0$  unless  $p^2 \mid n(n+1)$  (Proposition 3.6).
- (5) The integer  $J = g_1 - g_0 g_{d-1} = g_0 \sum_i (\alpha_i - \alpha_i^{-1})$  satisfies  $0 < |J| < 2.01 a \ln a$  (Lemma 2.4) and  $\text{rad}(b(n+1)) \mid J$  (Proposition 3.8). This restricts the primes that can enter step 4.

Steps 2–5 are incompatible when  $a, b \geq 5$  (Section 5). The case  $\min(a, b) \leq 4$  is excluded by step 5 alone (Section 6).

**Notation.** *General.* Throughout,  $n \geq 1000$  and

$$F(x) = (x-1)^2 f(x) = x^{n+1} - (n+1)x + n;$$

$\ln$  is the natural logarithm. For a positive integer  $k$ ,  $\text{rad}(k)$  is the product of the distinct primes dividing  $k$ . For a prime  $p$ ,  $v_p$  is the  $p$ -adic valuation on  $\mathbb{Q}$ . If  $P = \prod_{i=1}^d (x - \alpha_i)$  and  $P_2$  are monic,  $\text{Disc}(P) = \prod_{i < j} (\alpha_i - \alpha_j)^2$  and  $\text{Res}(P, P_2) = \prod_i P_2(\alpha_i)$ . The multiplicativity  $\text{Disc}(P_1 P_2) = \text{Disc}(P_1) \text{Disc}(P_2) \text{Res}(P_1, P_2)^2$  and the integrality of discriminants and resultants of monic integer polynomials will be used freely.

*Factorizations.* By Gauss's lemma, if  $f$  is reducible over  $\mathbb{Q}$  then  $f = gh$  with  $g, h \in \mathbb{Z}[x]$  monic and nonconstant; we call such a pair a *factorization*. For a factorization we write

$$d = \deg g, \quad g_j = [x^j]g, \quad a = g_0 = g(0), \quad b = h(0),$$

and  $\alpha_1, \dots, \alpha_d$  for the roots of  $g$ . We also use three derived quantities:

- the integer  $J = g_1 - g_0 g_{d-1}$ , which is small and nonzero (Lemma 2.4) and divisible by every prime factor of  $b(n+1)$  (Proposition 3.8);
- the product  $Q = \prod_{p^2 \mid n(n+1)} p$  of the only primes that can divide  $\text{Res}(g, h)$  (Proposition 3.6, Corollary 3.7);
- the real number

$$G = \frac{d}{n} (\ln n - \ln a + \ln \ln(2n+1)) - \frac{d^2}{2n^2} \ln(2n+1),$$

which controls the lower bound for  $|\text{Res}(g, h)|$  in Section 4 (Corollary 4.2).

The roles of  $g$  and  $h$  are symmetric until Section 5.

2. THE ROOTS OF  $f$  AND THE SHAPE OF A FACTORIZATION

Direct multiplication gives

$$(2.1) \quad (x-1)f(x) = \sum_{j=1}^n x^j - n, \quad F(x) = (x-1)^2 f(x) = x^{n+1} - (n+1)x + n,$$

and  $f(1) = n(n+1)/2 \neq 0$ .

**Lemma 2.1** (Annulus). *Every root  $\alpha$  of  $f$  satisfies  $1 < |\alpha| < (2n+1)^{1/n}$  and  $\alpha \notin [0, \infty)$ .*

*Proof.* The coefficients of  $f$  are positive, so  $f$  has no root in  $[0, \infty)$ . If  $|\alpha| \leq 1$ , then by (2.1)  $n = |\sum_{j=1}^n \alpha^j| \leq \sum_{j=1}^n |\alpha|^j \leq n$ . Equality forces every  $\alpha^j$  to be a nonnegative real number, in particular  $\alpha \geq 0$ , which is impossible. From  $\alpha^{n+1} = (n+1)\alpha - n$  we get

$$(2.2) \quad \alpha^n = n+1 - n/\alpha,$$

so  $|\alpha|^n \leq n+1 + n/|\alpha| < 2n+1$ .  $\square$

**Lemma 2.2** (Derivative at a root). *Every root  $\alpha$  of  $f$  satisfies  $F'(\alpha) = n(n+1)(\alpha-1)/\alpha$  and  $f'(\alpha) = n(n+1)/(\alpha(\alpha-1))$ . In particular  $f$  is separable.*

*Proof.* Here  $F'(x) = (n+1)(x^n - 1)$ , and (2.2) gives  $F'(\alpha) = (n+1)(n - n/\alpha)$ . Differentiating  $F = (x-1)^2 f$  at  $\alpha$  gives  $F'(\alpha) = (\alpha-1)^2 f'(\alpha)$ , and  $\alpha \neq 0, 1$  by Lemma 2.1.  $\square$

**Lemma 2.3.** *For every factorization:*

- (a)  $g(t) > 0$  and  $h(t) > 0$  for  $t \geq 0$ ; in particular  $a, b \geq 2$  and  $g(1), h(1) \geq 1$ , with  $g(1)h(1) = n(n+1)/2$ .
- (b)  $ab = n$  and  $\gcd(a, b) = \gcd(a, g_1) = 1$ ; hence  $\gcd(a, b(n+1)) = 1$ .

*Proof.* (a)  $g$  is monic without roots in  $[0, \infty)$ , so it is positive there. Moreover  $a = |\prod_i \alpha_i| > 1$  by Lemma 2.1. (b) Comparing the constant and linear coefficients of  $f = gh$  gives  $ab = n$  and  $ah_1 + bg_1 = n-1$ . A common prime factor of  $a$  and  $b$ , or of  $a$  and  $g_1$ , would divide  $n$  and  $n-1$ .  $\square$

**Lemma 2.4.** *For every factorization:*

- (a)  $J = a \sum_{i=1}^d (\alpha_i - \alpha_i^{-1})$ ,  $\gcd(a, J) = 1$ , and  $J \neq 0$ .
- (b)  $|J| < 2.01 a \ln a$ .

*Proof.* (a) From  $g = \prod (x - \alpha_i)$  we get  $g_{d-1} = -\sum \alpha_i$  and  $g_1 = -g_0 \sum \alpha_i^{-1}$ , which gives the formula since  $g_0 = a$ . Since  $J \equiv g_1 \pmod{a}$ , Lemma 2.3(b) gives  $\gcd(a, J) = 1$ , and  $a \geq 2$  forces  $J \neq 0$ .

(b) Complex conjugation permutes the  $\alpha_i$ , so  $\sum \alpha_i^{-1} = \sum \bar{\alpha}_i^{-1}$ . Moreover  $\alpha - \bar{\alpha}^{-1} = (|\alpha| - |\alpha|^{-1})\alpha/|\alpha|$  has modulus  $2 \sinh t$  with  $t = \ln |\alpha|$ . Put  $\tau = \ln(2n+1)/n$ ; by Lemma 2.1,  $0 < t_i := \ln |\alpha_i| < \tau$ . Since  $\sinh t/t$  is increasing and  $\sum t_i = \ln a$ ,

$$|J| \leq a \sum_i 2 \sinh t_i \leq 2 \frac{\sinh \tau}{\tau} a \ln a.$$

Comparing power series gives  $\sinh t/t \leq (1 - t^2/6)^{-1}$  for  $0 < t < \sqrt{6}$ . Since  $n \geq 1000$ , we have  $\tau < 1/100$ , whence  $2 \sinh \tau/\tau \leq 120000/59999 < 2.01$ .  $\square$

**Lemma 2.5.** *For every factorization,*

$$(2.3) \quad |\text{Disc}(g)| \cdot |\text{Res}(g, h)| \cdot g(0) \cdot g(1) = (n(n+1))^d.$$

*In particular, every prime dividing  $\text{Res}(g, h)$  divides  $n(n+1)$ .*

*Proof.* Multiplying Lemma 2.2 over the roots of  $g$  gives  $\prod_i f'(\alpha_i) \cdot \prod_i \alpha_i \cdot \prod_i (\alpha_i - 1) = (n(n+1))^d$ . Here  $\prod_i \alpha_i = (-1)^d g(0)$  and  $\prod_i (\alpha_i - 1) = (-1)^d g(1)$ , while  $f'(\alpha_i) = g'(\alpha_i)h(\alpha_i)$  and  $\prod_i g'(\alpha_i) = (-1)^{d(d-1)/2} \text{Disc}(g)$ . Take absolute values. All four factors in (2.3) are nonzero integers, and  $g(0), g(1) > 0$  by Lemma 2.3(a).  $\square$

### 3. LOCAL STRUCTURE OF THE ROOTS

This section works one prime at a time. Subsection 3.1 collects the standard facts from local field theory that we use. Subsection 3.2 locates the roots of  $f$  in  $\overline{\mathbb{Q}_p}$ , and Subsection 3.3 draws the consequences for a factorization  $f = gh$ . Only the results of Subsection 3.3 are used outside this section.

**3.1. Background from local fields.** Let  $p$  be a prime and  $m \geq 1$  an integer with  $p \nmid m$ . The following facts are standard; see [3, Ch. II] or [2, Ch. 7].

*Valuations.* Fix an algebraic closure  $\overline{\mathbb{Q}_p}$ , and let  $v$  be the valuation on it normalized by  $v(p) = 1$ ; on  $\mathbb{Q}$  it restricts to  $v_p$ . It satisfies  $v(xy) = v(x) + v(y)$  and  $v(x + y) \geq \min\{v(x), v(y)\}$ , with equality when  $v(x) \neq v(y)$ . Let  $\mathcal{O} = \{x : v(x) \geq 0\}$  be its valuation ring. Two elements of  $\mathcal{O}$  have the same image in the residue field exactly when their difference has positive valuation. If  $P \in \mathcal{O}[x]$  is monic, then all its roots lie in  $\mathcal{O}$ , and the number of roots with a given residue, counted with multiplicity, equals the multiplicity of that residue as a root of the reduction  $\overline{P}$ .

*An unramified extension.* The field  $E = \mathbb{Q}_p(\mu_m)$  generated by the  $m$ -th roots of unity is unramified over  $\mathbb{Q}_p$ ; in particular  $v(E^\times) = \mathbb{Z}$ . Since  $p \nmid m$ , the polynomial  $x^m - 1$  is separable modulo  $p$ , so distinct elements of  $\mu_m$  have distinct residues.

*Newton polygons.* Let  $P(y) = \sum_j c_j y^j \in E[y]$  with  $c_0 \neq 0$ , and take the lower convex hull of the points  $(j, v(c_j))$  with  $c_j \neq 0$ . If it has a segment of horizontal length  $\ell$  and slope  $-\lambda$ , then  $P$  has exactly  $\ell$  roots of valuation  $\lambda$ , counted with multiplicity, and the monic polynomial with these roots lies in  $E[y]$ .

The only consequence we derive in this generality is the following.

**Lemma 3.1** (Denominator criterion). *Let  $\zeta \in E$ , let  $e \geq 1$  be an integer, and let  $\varphi \in E[x]$  be monic with  $v(\alpha - \zeta) = 1/e$  for all its roots  $\alpha$ . Then  $e$  divides the degree of every monic divisor of  $\varphi$  in  $E[x]$ . In particular,  $\varphi$  is irreducible over  $E$  if  $\deg \varphi = e$ .*

*Proof.* If  $\psi \mid \varphi$  is monic, then  $v(\psi(\zeta)) = \deg \psi / e$ , and this lies in  $v(E^\times) = \mathbb{Z}$ .  $\square$

**3.2. The roots of  $f$  over  $\overline{\mathbb{Q}_p}$ .** From now on in this section, fix a prime  $p$  dividing  $n(n+1)$ . Since  $\gcd(n, n+1) = 1$ , exactly one of  $n, n+1$  is divisible by  $p$ ; write it as  $p^r m$  with  $r \geq 1$  and  $p \nmid m$ , and let  $E = \mathbb{Q}_p(\mu_m)$  and  $v$  be as in Subsection 3.1.

**Lemma 3.2.** *Every root of  $F$  lies in  $\mathcal{O}$ , and  $\overline{F} = x^\varepsilon (x^m - 1)^{p^r}$ , where  $\varepsilon = 1$  if  $p \mid n$  and  $\varepsilon = 0$  if  $p \mid n+1$ . Consequently every root of  $f$  is congruent to 0 or to an element of  $\mu_m$ , and for  $\zeta \in \mu_m$  the class of  $\zeta$ , that is, the set of roots  $\alpha$  of  $f$  with  $v(\alpha - \zeta) > 0$ , has  $p^r$  elements if  $\zeta \neq 1$  and  $p^r - 2$  if  $\zeta = 1$ . The class of 0 is empty if  $p \mid n+1$ ; if  $p \mid n$  it consists of a single root, of valuation  $r$ . Roots in different classes have difference of valuation 0.*

*Proof.*  $F$  is monic with coefficients in  $\mathbb{Z}_p$ , so its roots are integral. Modulo  $p$  we have  $F \equiv x^{n+1} - 1 \equiv (x^m - 1)^{p^r}$  if  $p \mid n+1$ , and  $F \equiv x^{n+1} - x = x(x^n - 1) \equiv x(x^m - 1)^{p^r}$  if  $p \mid n$ . Since  $F = \prod (x - \alpha)$  over  $\mathcal{O}$ , the number of roots with a given residue is the multiplicity of that residue in  $\overline{F}$ . Removing the double root 1 of  $F$  gives the counts for  $f$ . If  $p \mid n$ , the Newton polygon of  $F$  has the segment from  $(0, r)$  to  $(1, 0)$ , since  $F(0) = n$  and  $F'(0) = -(n+1)$  is a unit; this gives one root of valuation  $r$ .  $\square$

**Lemma 3.3.** *If  $p \mid n+1$ , then  $v_p\binom{n+1}{j} = r - v_p(j)$  for  $1 \leq j \leq p^r$ . If  $p \nmid n$ , then  $v_p\binom{n+1}{j} = r - v_p(j(j-1))$  for  $2 \leq j \leq p^r + 1$ .*

*Proof.* For  $k < p^r$  the binomial coefficient  $\binom{p^r m - 1}{k} = \prod_{i=1}^k (p^r m/i - 1)$  is congruent to  $(-1)^k$  modulo  $p$ , because  $v_p(p^r m/i) = r - v_p(i) \geq 1$ . Now use  $\binom{n+1}{j} = \frac{n+1}{j} \binom{n}{j-1}$  in the first case and  $\binom{n+1}{j} = \frac{n(n+1)}{j(j-1)} \binom{n-1}{j-2}$  in the second.  $\square$

**Proposition 3.4** (Packets). *Let  $\zeta \in \mu_m$ . The class of  $\zeta$  is a disjoint union of sets  $P_1, \dots, P_t$  (packets) with  $|P_j| = s_j$  and  $v(\alpha - \zeta) = \rho_j$  for  $\alpha \in P_j$ , where  $(s_j)$  and  $(\rho_j)$  are given in Table 1. In every case  $\rho_1 > \rho_2 > \dots > \rho_t$ . Each  $P_j$  is the set of roots of a monic  $\varphi_j \in E[x]$ , and  $\varphi_j$  is irreducible over  $E$ , with one exception: when  $p = 2$ ,  $\zeta \neq 1$  and  $r \geq 2$ , the quartic  $\varphi_1$  may be reducible, but all its monic divisors in  $E[x]$  have even degree.*

| case                                  | sizes $s_1, \dots, s_t$                      | radii $\rho_j$                  |
|---------------------------------------|----------------------------------------------|---------------------------------|
| $p$ odd, $\zeta \neq 1$               | $p, p(p-1), p^2(p-1), \dots, p^{r-1}(p-1)$   | $1/s_j$                         |
| $p$ odd, $\zeta = 1$                  | $p-2, p(p-1), p^2(p-1), \dots, p^{r-1}(p-1)$ | $1/s_j$                         |
| $p = 2$ , $\zeta \neq 1$ , $r = 1$    | 2                                            | $1/2$                           |
| $p = 2$ , $\zeta \neq 1$ , $r \geq 2$ | $4, 4, 8, \dots, 2^{r-1}$                    | $1/2, 1/4, 1/8, \dots, 2^{1-r}$ |
| $p = 2$ , $\zeta = 1$ , $r \geq 2$    | $2, 4, \dots, 2^{r-1}$                       | $1/s_j$                         |

TABLE 1. Packet sizes and radii. The odd-prime lists contain  $r$  packets; for  $r = 1$  only the first entry is present. The binary lists with  $r \geq 2$  contain  $r - 1$  packets; thus for  $r = 2$  only the initial quartic ( $\zeta \neq 1$ ) or quadratic ( $\zeta = 1$ ) is present. For  $p = 2$ ,  $\zeta = 1$ ,  $r = 1$  the class is empty. For  $p = 3$ ,  $\zeta = 1$ , the first packet has size 1.

*Proof.* First let  $\zeta \neq 1$  and  $\Psi(y) = F(\zeta + y) = \sum_j c_j y^j$ . By Lemma 3.2, the roots of  $\Psi$  of positive valuation correspond exactly to the class of  $\zeta$ . We have  $c_0 = F(\zeta) = e_0(1 - \zeta)$ , where  $e_0 \in \{n, n+1\}$  is the member divisible by  $p$  (use  $\zeta^{n+1} = 1$ , resp.  $\zeta^{n+1} = \zeta$ ). Next  $c_1 = F'(\zeta) = (n+1)(\zeta^n - 1)$ , which equals  $(n+1)(\zeta^{-1} - 1)$  if  $p \mid n+1$  and 0 if  $p \nmid n$ . Finally  $c_j = \binom{n+1}{j} \zeta^{n+1-j}$  for  $j \geq 2$ . Since  $1 - \zeta$  is a unit,  $v(c_0) = r$ . By Lemma 3.3, for  $1 \leq j \leq p^r$  with  $p^k \leq j < p^{k+1}$  we have  $v(c_j) \geq r - k$ , with equality when  $j = p^k$  and  $k \geq 1$ . The slopes  $-1/p$  and  $-1/(p^{k+1} - p^k)$ ,  $k \geq 1$ , are nondecreasing. Hence, on  $[0, p^r]$ , the Newton polygon has vertices  $(0, r)$  and  $(p^k, r - k)$  for  $1 \leq k \leq r$ , except that for  $p = 2$  and  $r \geq 2$  the points  $(0, r)$ ,  $(2, r-1)$ ,  $(4, r-2)$  are collinear and form a single segment. Reading off lengths and slopes gives the rows of Table 1 with  $\zeta \neq 1$ .

For  $\zeta = 1$  use  $f(1+y) = y^{-2}F(1+y) = \sum_{k \geq 0} \binom{n+1}{k+2} y^k$ . Lemma 3.3 gives the vertices  $(0, r)$  and  $(p^k - 2, r - k)$  for  $1 \leq k \leq r$  when  $p$  is odd. When  $p = 2$  it gives the vertices  $(p^k - 2, r - k)$  for  $1 \leq k \leq r$ , the first being  $(0, r-1)$ . All other points lie on or above the resulting polygons, by the same argument as before. This gives the rows with  $\zeta = 1$ .

In all rows with  $\rho_j = 1/s_j$ , Lemma 3.1 gives the irreducibility of  $\varphi_j$ . For the binary quartic,  $\rho_1 = 1/2$ , and Lemma 3.1 gives the evenness of the degrees of its divisors.  $\square$

**3.3. Consequences for a factorization.** We now bring in a factorization  $f = gh$ . Its effect on  $p$ -adic distances is governed by the packets of Proposition 3.4: a factor defined over  $\mathbb{Q}$  takes whole irreducible packets. The only possible exception is a binary quartic packet, which can split

into two quadratic factors. We first isolate the combinatorial fact used to bound the resulting cuts.

**Lemma 3.5** (A max-cut bound). *Let  $w_1, \dots, w_s \geq 0$  satisfy  $w_k \geq \sum_{i < k} w_i$  for  $2 \leq k \leq s-1$ . Give the edge  $\{i, j\}$ ,  $i < j$ , of the complete graph on  $\{1, \dots, s\}$  the weight  $w_i$ . Then every cut has weight at most  $M(w_1, \dots, w_s)$ , where  $M = 0$  for  $s \leq 1$  and*

$$M(w_1, \dots, w_s) = \sum_{i < s} w_i + M(w_1, \dots, w_{s-2}) \quad (s \geq 2).$$

*Proof.* Induct on  $s$ . Take a cut of maximal weight. If  $s-1$  and  $s$  lie on the same side, moving  $s-1$  to the other side gains  $w_{s-1}$  from the edge  $\{s-1, s\}$  and loses at most  $\sum_{i < s-1} w_i \leq w_{s-1}$ . So we may assume that  $s-1$  and  $s$  are separated. Then each  $i < s-1$  contributes exactly  $w_i$  through its two edges to  $\{s-1, s\}$ , the edge  $\{s-1, s\}$  contributes  $w_{s-1}$ , and the remaining edges form a cut of the instance  $(w_1, \dots, w_{s-2})$ , which still satisfies the hypothesis.  $\square$

**Proposition 3.6** (Local resultant bound). *Let  $r = v_p(n(n+1))$ . For every factorization,*

$$(3.1) \quad v_p(\text{Res}(g, h)) \leq (n+1) \sum_{1 \leq j \leq r/2} p^{1-2j} - 2\lfloor r/2 \rfloor.$$

*In particular  $v_p(\text{Res}(g, h)) = 0$  if  $p^2 \nmid n(n+1)$ .*

*Proof.* Put  $C = \sum_{1 \leq j \leq r/2} p^{r+1-2j}$ , so that  $C = 0$  if  $r \leq 1$ . We show that each residue class contributes at most  $C$ , and the class of 1 at most  $C - 2\lfloor r/2 \rfloor$ . Summing over the  $m$  classes then gives  $v_p(\text{Res}(g, h)) \leq mC - 2\lfloor r/2 \rfloor$ , and  $mC = mp^r \sum_{1 \leq j \leq r/2} p^{1-2j}$  with  $mp^r \leq n+1$ , which is (3.1).

We have  $v_p(\text{Res}(g, h)) = \sum v(\alpha - \beta)$ , the sum over roots  $\alpha$  of  $g$  and  $\beta$  of  $h$ . By Lemma 3.2, only pairs in the same class  $\zeta \in \mu_m$  contribute. Since  $g, h \in E[x]$  and  $f$  is separable, each  $E$ -irreducible factor of  $f$  divides exactly one of  $g, h$ . Hence each packet lies entirely in the roots of  $g$  or of  $h$ , except that a binary quartic may be split into two quadratics, one in each.

If  $\alpha \in P_i$  and  $\beta \in P_j$  with  $i < j$ , then  $\rho_i > \rho_j$  gives  $v(\alpha - \beta) = \rho_j$ . The total over such pairs is  $s_i s_j \rho_j$ , which equals  $s_i$  whenever  $\rho_j = 1/s_j$ .

Odd  $p$ ,  $r \geq 2$ . In the class of  $\zeta$ , the contribution is the weight of the cut, induced by the assignment of packets to  $g$  and  $h$ , with weights  $w_i = s_i$ . These weights satisfy the hypothesis of Lemma 3.5. For  $\zeta \neq 1$  we have  $\sum_{i < r} s_i = p^{r-1}$ , and for  $\zeta = 1$  it is  $p^{r-1} - 2$ . Hence the maximum is  $C$ , resp.  $C - 2\lfloor r/2 \rfloor$ .

$p = 2$ ,  $\zeta = 1$ ,  $r \geq 2$ . Prepend a dummy vertex of weight 0. Then the weights  $0, 2, 4, \dots, 2^{r-1}$  satisfy the hypothesis,  $\sum_{i < r} w_i = 2^{r-1} - 2$ , and the maximum is  $C - 2\lfloor r/2 \rfloor$ .

$p = 2$ ,  $\zeta \neq 1$ ,  $r \geq 2$ . We first bound the contribution of a split quartic  $\varphi_1 = \varphi' \varphi''$ . For a root  $\alpha$  of  $\varphi_1$ , Lemma 2.2 gives  $v(F'(\alpha)) = v(n(n+1)) = r$ , since  $\alpha$  and  $\alpha - 1$  are units. On the other hand  $v(F'(\alpha)) = \sum_{\beta \neq \alpha} v(\alpha - \beta)$  over the roots of  $F$ , counted with multiplicity. In particular, the double root 1 appears twice and contributes zero, since  $\alpha - 1$  is a unit. The other roots of  $\varphi_1$  contribute  $v(\varphi'_1(\alpha))$ , each packet  $P_j$  with  $j \geq 2$  contributes  $s_j \rho_j = 1$ , and roots in other classes contribute 0. Hence  $v(\varphi'_1(\alpha)) = 2$  and  $v(\text{Disc } \varphi_1) = 8$ . Translate by  $\zeta$  and write  $\varphi'(\zeta + y) = y^2 + c_1 y + c_0$ . Both roots have valuation  $1/2$ , so  $v(c_0) = 1$  and  $v(c_1) \geq 1/2$ ; since  $c_1 \in E$ , in fact  $v(c_1) \geq 1$ . Thus  $v(\text{Disc } \varphi') = v(c_1^2 - 4c_0) \geq 2$ , and likewise for  $\varphi''$ . From  $\text{Disc } \varphi_1 = \text{Disc } \varphi' \text{Disc } \varphi'' \text{Res}(\varphi', \varphi'')^2$  we get  $v(\text{Res}(\varphi', \varphi'')) \leq 2$ .

Now model the class by  $r$  vertices with weights  $2, 2, 4, 8, \dots, 2^{r-1}$ . The first two vertices represent the two halves of  $\varphi_1$ , placed on the same side when  $\varphi_1$  is not split. Each half has 2 roots, so its pairs with  $P_j$ ,  $j \geq 2$ , contribute  $2s_j \rho_j = 2$ . The edge between the two halves has weight 2, which bounds the internal contribution just computed. The weights satisfy the hypothesis of Lemma 3.5,  $\sum_{i < r} w_i = 2^{r-1}$ , and the maximum is  $C$ .

$r = 1$ . Every class consists of at most one packet (or is empty), so no pair contributes, in agreement with  $C = 0$ .  $\square$

**Corollary 3.7.** *For every factorization:*

(a) *one has*

$$(3.2) \quad \ln |\operatorname{Res}(g, h)| \leq (n+1) \sum_{p|Q} \frac{p \ln p}{p^2 - 1}.$$

(b) *one has*

$$(3.3) \quad \ln |\operatorname{Res}(g, h)| < \left( \frac{n+1}{16} - 1 \right) \ln(n(n+1)) + \frac{13}{20}(n+1).$$

*Proof.* By Lemma 2.5,  $\ln |\operatorname{Res}(g, h)| = \sum_{p|n(n+1)} v_p(\operatorname{Res}(g, h)) \ln p$ , and by Proposition 3.6 only  $p \mid Q$  contribute, each with

$$v_p(\operatorname{Res}(g, h)) \leq (n+1) \sum_{1 \leq j \leq r/2} p^{1-2j} < (n+1) \frac{p}{p^2 - 1},$$

which gives (3.2). For (3.3), write  $p^{1-2j} = \frac{1}{8} + (p^{1-2j} - \frac{1}{8})$ . The bracket is positive only for  $(p, j) = (2, 1), (3, 1), (5, 1), (7, 1)$ , where it equals  $\frac{3}{8}, \frac{5}{24}, \frac{3}{40}$  and  $\frac{1}{56}$ . Hence

$$v_p(\operatorname{Res}(g, h)) \leq \left( \frac{n+1}{8} - 2 \right) \left\lfloor \frac{r}{2} \right\rfloor + (n+1) \kappa_p,$$

with  $\kappa_2 = \frac{3}{8}, \kappa_3 = \frac{5}{24}, \kappa_5 = \frac{3}{40}, \kappa_7 = \frac{1}{56}$ , and  $\kappa_p = 0$  otherwise. Since  $n+1 \geq 16$  and  $\sum_p [v_p(n(n+1))/2] \ln p \leq \frac{1}{2} \ln(n(n+1))$ ,

$$\ln |\operatorname{Res}(g, h)| \leq \left( \frac{n+1}{16} - 1 \right) \ln(n(n+1)) + (n+1) \left( \frac{3}{8} \ln 2 + \frac{5}{24} \ln 3 + \frac{3}{40} \ln 5 + \frac{1}{56} \ln 7 \right),$$

and the last bracket is less than  $0.645 < 13/20$ .  $\square$

**Proposition 3.8.** *For every factorization,  $\operatorname{rad}(b(n+1)) \mid J$ .*

*Proof.* Let  $p \mid b(n+1)$ , so  $p \nmid a$ . Every root of  $g$  is integral, and  $v(a) = \sum_i v(\alpha_i) = 0$ , so all roots of  $g$  are units and the root of the class of 0 (if any) belongs to  $h$ . Thus the roots of  $g$  lie in classes  $\zeta \in \mu_m$ . For  $\zeta = 1$ , each  $\alpha - \alpha^{-1}$  has positive valuation. For  $\zeta \neq 1$ , by the proof of Proposition 3.6, the roots of  $g$  in the class of  $\zeta$  form a union of sets whose sizes are divisible by  $p$ : packets of sizes listed in Table 1, or halves of a binary quartic. The sum of  $\alpha - \alpha^{-1}$  over a set of size  $s$  in the class of  $\zeta$  is congruent to  $s(\zeta - \zeta^{-1}) \equiv 0$  modulo the maximal ideal. Hence  $v(\sum_i (\alpha_i - \alpha_i^{-1})) > 0$ , so  $v_p(J) > 0$  by Lemma 2.4(a).  $\square$

#### 4. A LOWER BOUND FOR THE RESULTANT FROM THE DISCRIMINANT

Section 3 bounded  $|\operatorname{Res}(g, h)|$  from above, one prime at a time. Here we bound it from below, using ordinary complex absolute values. The roots of  $g$  all lie in the thin annulus of Lemma 2.1. This controls the product of their pairwise distances and gives an upper bound on  $|\operatorname{Disc}(g)|$  (Lemma 4.1). The identity of Lemma 2.5 then forces  $|\operatorname{Res}(g, h)|$  to be large (Corollary 4.2).

**Lemma 4.1.** *Let  $P \in \mathbb{C}[x]$  be monic of degree  $d \geq 1$  with distinct nonzero roots whose logarithmic moduli lie in an interval of length  $\delta > 0$ . Then*

$$\ln |\operatorname{Disc} P| \leq (d-1) \ln |P(0)| + \frac{\delta d^2}{2} - d \ln \delta.$$

*Proof.* Let  $\alpha_1, \dots, \alpha_d$  be the roots,  $u_i = \alpha_i/|\alpha_i|$  and  $t = e^{-\delta}$ . For  $i \neq j$  the ratio  $q = |\alpha_i|/|\alpha_j|$  lies in  $[t, t^{-1}]$ , so  $q + q^{-1} \leq t + t^{-1}$  and

$$\frac{|\alpha_i - \alpha_j|^2}{|\alpha_i||\alpha_j|} = q + q^{-1} - 2\operatorname{Re}(u_i \overline{u_j}) \leq t^{-1}|1 - tu_i \overline{u_j}|^2.$$

For  $|z| = 1$  and  $0 < t < 1$ ,  $\ln|1 - tz| = -\sum_{k \geq 1} \frac{t^k}{k} \operatorname{Re} z^k$ . Hence

$$\sum_{i \neq j} \ln|1 - tu_i \overline{u_j}| = -\sum_{k \geq 1} \frac{t^k}{k} \left| \sum_i u_i^k \right|^2 - d \ln(1 - t) \leq -d \ln(1 - t).$$

Summing the pairwise inequality over  $i < j$  gives

$$\ln |\operatorname{Disc} P| \leq (d-1) \ln |P(0)| + \frac{d(d-1)}{2} \delta - d \ln(1 - t).$$

Finally  $1 - e^{-\delta} \geq \delta e^{-\delta/2}$ , i.e.  $2 \sinh(\delta/2) \geq \delta$ . □

**Corollary 4.2.** *For every factorization,*

$$\ln |\operatorname{Res}(g, h)| \geq nG - \ln \frac{n(n+1)}{2}.$$

*Proof.* By Lemma 2.1, the numbers  $\ln |\alpha_i|$  lie in the interval  $(0, \ln(2n+1)/n)$ , so Lemma 4.1 applies with  $\delta = \ln(2n+1)/n$ . Taking logarithms in (2.3) gives

$$\begin{aligned} d \ln(n(n+1)) &\leq (d-1) \ln a + \frac{d^2}{2n} \ln(2n+1) + d \ln \frac{n}{\ln(2n+1)} \\ &\quad + \ln |\operatorname{Res}(g, h)| + \ln g(0) + \ln g(1). \end{aligned}$$

Since  $g(0) = a$ , this rearranges to  $\ln |\operatorname{Res}(g, h)| \geq nG + d \ln(1 + \frac{1}{n}) - \ln g(1)$ . Finally  $d \ln(1 + \frac{1}{n}) \geq 0$ , and  $g(1) \leq g(1)h(1) = n(n+1)/2$  by Lemma 2.3(a). □

## 5. FACTORIZATIONS WITH LARGE CONSTANT TERMS

In this section we label the factors so that  $d \leq (n-1)/2$ . Then Lemma 2.1 gives

$$(5.1) \quad 0 < \ln a < \frac{d}{n} \ln(2n+1) < \frac{1}{2} \ln(2n+1).$$

The three lemmas below bound the single quantity  $\ln |\operatorname{Res}(g, h)|$ , each in the form  $nX - \ln \frac{n(n+1)}{2}$  for an explicit  $X$ . Lemmas 5.1 and 5.2 are upper bounds, coming from the  $p$ -adic budgets of Corollary 3.7; Lemma 5.3 is a lower bound, coming from Corollary 4.2.

**Lemma 5.1.**  $\ln |\operatorname{Res}(g, h)| < n(\frac{1}{8} \ln n + \frac{33}{50}) - \ln \frac{n(n+1)}{2}$ .

*Proof.* By (3.3), it suffices to show  $\frac{n+1}{16} \ln(n(n+1)) + \frac{13}{20}(n+1) - \ln 2 \leq n(\frac{1}{8} \ln n + \frac{33}{50})$ . Using  $\ln(n(n+1)) < 2 \ln n + \frac{1}{n}$ , the left side is less than

$$n\left(\frac{1}{8} \ln n + \frac{13}{20}\right) + \frac{1}{8} \ln n + \frac{13}{20} + \frac{n+1}{16n} - \ln 2,$$

and the last four terms are at most  $n/100$ : their total is below 1 at  $n = 1000$  (using  $\ln 1000 < 7$ ) and grows more slowly than  $n/100$ . □

**Lemma 5.2.**  $\ln |\operatorname{Res}(g, h)| < n(\frac{1}{5} \ln a + 1.03) - \ln \frac{n(n+1)}{2}$ .



*Proof.* Let  $p \mid Q$ . If  $p \mid a$ , then  $p \nmid b(n+1)$  and  $p^2 \mid ab(n+1)$ , so  $p^2 \mid a$ . The product of such primes is therefore at most  $\sqrt{a}$ . All other primes dividing  $Q$  divide  $b(n+1)$ . By Proposition 3.8 and Lemma 2.4(b),

$$Q \leq \sqrt{a} \operatorname{rad}(b(n+1)) \leq \sqrt{a} |J| < 2.01 a^{3/2} \ln a.$$

For  $p \geq 11$  we have  $p/(p^2 - 1) < 1/10$ , and  $\sum_{p \leq 7} (\frac{p}{p^2-1} - \frac{1}{10}) \ln p < 0.959$ . Hence  $\Sigma := \sum_{p \mid Q} \frac{p \ln p}{p^2-1} < \frac{1}{10} \ln Q + 0.97$ . With  $u = \ln a$  and the tangent-line bound  $\ln u \leq \frac{u}{2} + \ln 2 - 1$ , we have  $\ln Q < \frac{3}{2}u + \ln 2.01 + \ln u < 2u + 0.4$ , so  $\Sigma < \frac{u}{5} + 1.01$ . By (3.2),  $\ln |\operatorname{Res}(g, h)| \leq (n+1)\Sigma < n(\frac{u}{5} + 1.01) + \Sigma$ , so it suffices to show  $\Sigma + \ln \frac{n(n+1)}{2} \leq \frac{n}{50}$ . Using  $Q^2 \leq n(n+1)$  and  $\ln(n(n+1)) < 2 \ln n + 1/n$ , the left side is less than  $2.1 \ln n + 1$ , and  $(2.1 \ln n + 1)/n$  is decreasing and less than  $1/50$  at  $n = 1000$ .  $\square$

**Lemma 5.3.** *If  $a \geq 5$ , then*

$$\ln |\operatorname{Res}(g, h)| > n \min \left\{ \frac{1}{5} \ln a + 1.03, \frac{1}{8} \ln n + 0.73 \right\} - \ln \frac{n(n+1)}{2}.$$

*Proof.* Write  $t = \ln n$ ,  $u = \ln a$  and  $L = \ln(2n+1)$ , so that  $u \geq \ln 5$  and, by (5.1),  $u < dL/n < L/2$ . By Corollary 4.2, it suffices to show  $G > \min \{ \frac{u}{5} + 1.03, \frac{t}{8} + 0.73 \}$ . We have  $t > 6.9$ ,  $0 < L - t = \ln(2 + 1/n) < 0.7$  and  $\ln L > 2$ . Regard  $G$  as a function of  $x = d/n \in (u/L, 1/2)$ . By (5.1), its derivative satisfies  $t - u + \ln L - Lx > \ln L - (L - t) > 0$ , so

$$G > H(u) := \frac{u}{L} \left( t + \ln L - \frac{3}{2}u \right).$$

$H$  is a concave quadratic in  $u$ . We use three lower bounds for it. First, for  $u \leq 3$  the map  $s \mapsto (s + 2 - 1.5u)/(s + 0.7)$  is increasing, so

$$(5.2) \quad H(u) \geq \frac{u(8.9 - 1.5u)}{7.6} \quad (u \leq 3).$$

Second, for  $u \leq t/4$  we have  $t - \frac{3}{2}u \geq \frac{5}{8}t$  and  $\ln L > 2 > \frac{5}{8}(L - t)$ , so  $H(u) \geq \frac{5}{8}u$ . Third, at  $u = L/2$ ,

$$H - \frac{1}{8}t = \frac{1}{2} \ln L - \frac{3}{8}(L - t) > 1 - 0.2625 > 0.73.$$

Put  $u_1 = \max(3, t/4)$ .

(i)  $\ln 5 \leq u \leq u_1$ . Here  $H(u) - \frac{u}{5} - 1.03$  is concave, so it suffices to check the endpoints. At  $u = \ln 5 \in (1.609, 1.61)$ , (5.2) gives  $H > 1.372 > \frac{u}{5} + 1.03$ . At  $u = u_1$ : if  $u_1 = 3$ , (5.2) gives  $H(3) > 1.73 > 1.63$ ; if  $u_1 = t/4 \geq 3$ , then  $H(u_1) - \frac{u_1}{5} - 1.03 \geq 0.425u_1 - 1.03 > 0$ . So  $G > \frac{u}{5} + 1.03$ .

(ii)  $u_1 \leq u \leq L/2$ . Here  $H$  is concave, so it suffices to check that  $H > \frac{t}{8} + 0.73$  at both endpoints. At  $u = L/2$  this was shown above. If  $u_1 = 3$  (so  $t \leq 12$ ), then  $H(3) - \frac{t}{8} \geq 3 - \frac{9.6}{t+0.7} - \frac{t}{8}$ , which is concave in  $t$  and takes the values  $\frac{1329}{1520} > 0.74$  and  $\frac{189}{254} > 0.74$  at  $t = 6.9$  and  $t = 12$ , respectively. If  $u_1 = t/4$  (so  $t \geq 12$ ), then

$$H - \frac{1}{8}t \geq \frac{t}{t+0.7} \left( \frac{t}{32} + \frac{33}{80} \right) \geq \frac{12}{12.7} \cdot 0.7875 > 0.74.$$

So  $G > \frac{t}{8} + 0.73$ .  $\square$

**Proposition 5.4.** *There is no factorization with  $a, b \geq 5$ .*

*Proof.* Suppose  $a, b \geq 5$ , and label the factors so that  $d \leq (n-1)/2$ . The lower bound of Lemma 5.3 exceeds the upper bound of Lemma 5.2 if the minimum there is  $\frac{1}{5} \ln a + 1.03$ , and that of Lemma 5.1 if it is  $\frac{1}{8} \ln n + 0.73$ , since  $0.73 > \frac{33}{50}$ . Either way we have a contradiction.  $\square$

## 6. FACTORIZATIONS WITH A SMALL CONSTANT TERM

**Lemma 6.1.** *Every factorization has  $a \geq 5$  and  $b \geq 5$ .*

*Proof.* By symmetry it suffices to show  $a \geq 5$ ; suppose  $a \leq 4$ . The integers  $a$ ,  $b$  and  $n+1$  are pairwise coprime (Lemma 2.3(b)), and  $b \geq 2$  (Lemma 2.3(a)). So  $b$  and  $n+1$  each have a prime factor, these two primes are distinct, and neither divides  $a$ . Their product divides  $\text{rad}(b(n+1))$ , and it is at least  $3 \cdot 5 = 15$  if  $a \in \{2, 4\}$  and at least  $2 \cdot 5 = 10$  if  $a = 3$ . On the other hand, Proposition 3.8 and Lemma 2.4 give

$$\text{rad}(b(n+1)) \leq |J| < 2.01 a \ln a,$$

and the right side is less than 3, 7 and 12 for  $a = 2, 3, 4$ . This is a contradiction.  $\square$

*Proof of Theorem 1.1.* For  $n < 1000$  see Appendix A; let  $n \geq 1000$  and suppose  $f$  is reducible over  $\mathbb{Q}$ . By Gauss's lemma there is a factorization  $f = gh$ . Lemma 6.1 gives  $a, b \geq 5$ , which is excluded by Proposition 5.4. Hence  $f$  is irreducible over  $\mathbb{Q}$ , and since it is monic, also over  $\mathbb{Z}$ .  $\square$

7. REMOVING THE COMPUTATION FOR  $n < 1000$ 

The main argument assumed  $n \geq 1000$ , and Appendix A checks the smaller cases by computer. This section explains what has to be added to cover every  $n \geq 2$  by hand. Nothing here is used elsewhere. In this section  $n \geq 2$  is arbitrary.

**7.1. What carries over.** All results of Sections 2–4 hold for every  $n \geq 2$  with unchanged proofs, with two exceptions, and two results of Section 5 extend with the same proofs:

- Corollary 3.7(b) needs  $n \geq 15$ , since its proof uses  $n+1 \geq 16$ .
- Lemma 2.4(b) needs  $n \geq 1000$ . For  $n \geq 6$  the same proof, now with  $\tau = \ln(2n+1)/n < 1/2$  and  $2 \sinh \tau / \tau \leq 48/23$ , gives

$$(7.1) \quad |J| < 2.1 a \ln a.$$

- Lemma 5.1 holds for  $n \geq 182$ : in its proof, the four leftover terms total less than  $0.7 < 182/100$  at  $n = 182$ .
- Corollary 4.2 holds for any narrower annulus. If all the numbers  $\ln |\alpha|$  lie in an interval of length  $W/n$ , where  $W > 0$ , the same proof with  $\delta = W/n$  gives

$$(7.2) \quad \ln |\text{Res}(g, h)| \geq nG_W - \ln \frac{n(n+1)}{2}, \quad G_W = \frac{d}{n} (\ln n - \ln a + \ln W) - \frac{Wd^2}{2n^2}.$$

The inequality (5.1) also holds for every  $n$  once  $d \leq (n-1)/2$ .

A factorization forces  $n = ab$  with coprime  $a, b \geq 2$  (Lemma 2.3). So  $n \geq 6$  and  $n$  is not a prime power; in particular  $f$  is irreducible for  $n \leq 5$ .

The plan follows an earlier version of the main argument. Without  $n \geq 1000$ , two things break. First, the lower bound of Section 5 is too weak for small constant terms, so the threshold between “small” and “large” constant terms moves from 5 to 13, and the small constant terms  $5 \leq a \leq 12$  need arithmetic arguments (Subsections 7.3–7.5). Second, if both constant terms are at least 13 then  $n \geq 182$ , and for  $182 \leq n < 1000$  the lower bound needs a narrower annulus (Subsection 7.6).

## 7.2. Inner root bounds.

**Lemma 7.1.** *Let  $\alpha = \rho e^{i\theta}$  be a root of  $f$ .*

- (a) *If  $0 < \theta < \pi$ , then  $\theta > 2\pi/n$ .*
- (b) *If  $n \geq 3$ , then  $\rho^n > 4$ .*
- (c) *If  $n \geq 8$ , then  $n \ln \rho > 11/6$ .*

*Proof.* (a) Since  $\rho > 1$  (Lemma 2.1), we have  $n/\rho < n+1$ , so by (2.2) the number  $\alpha^n = n+1 - (n/\rho)e^{-i\theta}$  has positive real and imaginary parts. Hence  $2\pi k < n\theta < 2\pi k + \pi/2$  for some integer  $k \geq 0$ . If  $k = 0$ , then  $\text{Im } \alpha^j > 0$  for  $1 \leq j \leq n$ , contradicting  $\sum_{j=1}^n \alpha^j = n$ .

(b) A negative real root satisfies  $\rho^n = n+1 + n/\rho > n+1 \geq 4$  by (2.2); by conjugation we may assume  $0 < \theta < \pi$ . If  $\theta \geq \pi/2$ , then  $\text{Re } \alpha^n \geq n+1$  and  $\text{Im } \alpha^n > 0$ , so  $\rho^n > n+1$ . If  $\theta < \pi/2$ , then

$$\rho^{2n} = \left( \frac{n}{\rho} - (n+1) \cos \theta \right)^2 + (n+1)^2 \sin^2 \theta \geq (n+1)^2 \sin^2 \theta,$$

so by (a),  $\rho^n \geq (n+1) \sin \theta > (n+1) \cdot 2\theta/\pi > 4(n+1)/n > 4$ .

(c) In the case  $\theta < \pi/2$  of (b), (a) gives  $\sin \theta > \sin(2\pi/n)$ , and  $\sin t \geq t - t^3/6$  gives

$$\rho^n > 2\pi \frac{n+1}{n} \left( 1 - \frac{2\pi^2}{3n^2} \right) \geq 2\pi,$$

because  $n^2/(n+1) \geq 64/9 > 2\pi^2/3$ . In the other cases of (b),  $\rho^n > n+1 \geq 9 > 2\pi$ . Finally  $2\pi > e^{11/6}$ .  $\square$

## 7.3. Congruences and parity.

**Lemma 7.2.** *For every factorization:*

- (a)  *$nd$  is even;*
- (b)  *$a^{n+1} = (n+1)^d g(\frac{n}{n+1}) > (n(n+1))^{d/2}$ ;*
- (c)  *$a^{n+1} \equiv (-1)^d \pmod{n+1}$ ;*
- (d)  *$a^n \equiv 1 \pmod{b}$ .*

*Proof.* (a) If  $n$  is odd, then  $n+1$  is even and  $F(x) = x^{n+1} + (n+1)|x| + n > 0$  for  $x < 0$ . So  $f$ , and hence  $g$ , has no real roots, and  $d$  is even.

(b) Multiply  $\alpha^{n+1} = (n+1)(\alpha - \frac{n}{n+1})$  over the roots of  $g$ . This gives  $(-1)^{d(n+1)} a^{n+1} = (-1)^d (n+1)^d g(\frac{n}{n+1})$ , and  $(-1)^{dn} = 1$  by (a). For  $0 < t < 1$  and  $|z| > 1$  one has  $|z - t|^2 - t|z - 1|^2 = (1-t)(|z|^2 - t) > 0$ . Taking  $t = n/(n+1)$  and multiplying over the roots of  $g$  gives  $g(\frac{n}{n+1}) > (\frac{n}{n+1})^{d/2} g(1) \geq (\frac{n}{n+1})^{d/2}$ .

(c) Expanding  $(n+1)^d g(\frac{n}{n+1}) = \sum_j g_j n^j (n+1)^{d-j}$  gives  $a^{n+1} \equiv n^d \equiv (-1)^d \pmod{n+1}$ .

(d) Let  $C \in M_d(\mathbb{Z})$  be the companion matrix of  $g$ . Then  $g(C) = 0$ , so  $F(C) = 0$ , and  $\det C = (-1)^d a$ . Reducing  $C^{n+1} - (n+1)C + nI = 0$  modulo  $b$ , where  $n \equiv 0$  and  $n+1 \equiv 1$ , gives  $C^{n+1} \equiv C$ . Since  $\gcd(a, b) = 1$ ,  $C$  is invertible modulo  $b$ , so  $C^n \equiv I \pmod{b}$ . Taking determinants gives  $(-1)^{dn} a^n \equiv 1$ , and  $dn$  is even by (a).  $\square$

**Corollary 7.3.** *For every factorization,  $d$  is odd if  $2 \mid a$ , and even if  $2 \nmid a$ .*

*Proof.* Take  $p = 2$  in Section 3. By Table 1 and the proof of Proposition 3.6,  $g$  has an even number of roots in the unit classes. By Lemma 3.2, the class of 0 contains one root if  $2 \mid n$  and none otherwise, and this root belongs to  $g$  if and only if  $v(a) > 0$ , i.e.  $2 \mid a$ .  $\square$

#### 7.4. $n + 1$ a prime power.

**Lemma 7.4.** *Suppose  $n + 1 = p^r$  and let  $g$  be the factor of smaller degree in a factorization. Then  $|\text{Res}(g, \Phi_{n+1})| = p^d$ , where  $\Phi_{n+1}$  is the  $(n + 1)$ -th cyclotomic polynomial.*

*Proof.* Let  $v$  be the valuation on  $\overline{\mathbb{Q}_p}$  from Subsection 3.1. Here  $d \leq (n - 1)/2 < \varphi(n + 1)$ . Modulo  $p$  we have  $f \equiv (x - 1)^{n-1}$ , hence  $g \equiv (x - 1)^d$ . All nonleading coefficients of  $g(1 + y)$  are then divisible by  $p$ , so every root  $\alpha$  of  $g$  satisfies  $v(\alpha - 1) \geq 1/d$ . Every primitive  $(n + 1)$ -th root of unity  $\zeta$  satisfies  $v(\zeta - 1) = 1/\varphi(n + 1) < 1/d$ , since  $\Phi_{n+1}(1 + y)$  is  $p$ -Eisenstein. Hence  $v(\alpha - \zeta) = 1/\varphi(n + 1)$ ,  $v(\Phi_{n+1}(\alpha)) = 1$ , and  $v_p(\text{Res}(g, \Phi_{n+1})) = d$ .

At a primitive root  $\zeta$ , (2.1) gives  $f(\zeta) = (n + 1)(1 - \zeta)/(\zeta - 1)^2 = (n + 1)/(1 - \zeta)$ . Hence  $|\text{Res}(f, \Phi_{n+1})| = (n + 1)^{\varphi(n+1)}/\Phi_{n+1}(1) = (n + 1)^{\varphi(n+1)}/p$ , a power of  $p$ . Since  $\text{Res}(g, \Phi_{n+1})$  divides it, the claim follows.  $\square$

**Proposition 7.5.** *If  $n + 1$  is a prime power, then  $f$  is irreducible.*

*Proof.* Let  $n + 1 = p^r$  and suppose a factorization exists; let  $g$  have the smaller degree. Since  $n$  is not a prime power,  $n + 1 \notin \{3, 4, 5, 8\}$ .

$p$  odd,  $r \geq 3$ . Here  $n \geq 26$ , and Lemma 7.1(b) gives  $|\alpha|^{n+1} > 4$ . Hence

$$|\Phi_{n+1}(\alpha)| \geq \frac{|\alpha|^{n+1} - 1}{|\alpha|^{(n+1)/p} + 1} > \frac{3}{8}|\alpha|^{\varphi(n+1)}.$$

Multiplying over the roots of  $g$  and using Lemma 7.4 and Lemma 7.2(b),

$$p^d > \left(\frac{3}{8}\right)^d a^{\varphi(n+1)} > \left(\frac{3}{8}\right)^d (n(n + 1))^{d(p-1)/(2p)}.$$

But  $\sqrt{n(n + 1)} > n \geq p^3 - 1 > \frac{8}{9}p^3$ , so  $(n(n + 1))^{(p-1)/(2p)} > \frac{8}{9}p^{3(p-1)/p} \geq \frac{8}{9}p^2 \geq \frac{8}{3}p$ . This gives  $p > p$ , a contradiction.

$p = 2$ . Here  $\Phi_{n+1} = x^{(n+1)/2} + 1$  and  $|\alpha|^{(n+1)/2} > |\alpha|^{n/2} > 2$ , so  $|\Phi_{n+1}(\alpha)| > \frac{1}{2}|\alpha|^{(n+1)/2}$ . As above,  $2^d > 2^{-d}a^{(n+1)/2} > 2^{-d}(n(n + 1))^{d/4}$ , so  $n(n + 1) < 256$  and  $n + 1 \leq 16$ . The only remaining case is  $n + 1 = 16$ ,  $n = 15$ , with constant terms  $\{3, 5\}$ . Taking the factor with constant term 5 as  $g$ , Lemma 7.2(d) gives  $5^{15} \equiv 1 \pmod{3}$ , which is false.

$r = 2$ ,  $p$  odd. The degrees of  $g$  and  $h$  sum to  $p^2 - 2$ , which is odd, so exactly one of them is even. By Lemma 7.2(c) and  $a^{p^2} \equiv a \pmod{p}$ , the constant term of the even-degree factor is  $\equiv 1 \pmod{p}$ , and that of the odd-degree factor is  $\equiv -1 \pmod{p}$ . Being at least 2, they are at least  $p + 1$  and  $p - 1$  respectively, and their product is  $p^2 - 1$ . So they equal  $p + 1$  and  $p - 1$ , which are both even, contradicting  $\gcd(a, b) = 1$ .

$r = 1$ . Then  $f(1 + y) = \sum_k \binom{p}{k+2} y^k$  is  $p$ -Eisenstein, since its constant term is  $p(p - 1)/2$ .  $\square$

#### 7.5. Constant terms at most 12.

**Lemma 7.6.** *In every factorization, if  $a \leq 12$  then  $a = 11$ .*

*Proof.* The integers  $a, b, n + 1$  are pairwise coprime. By Proposition 7.5,  $n + 1$  has at least two prime factors, and  $b \geq 2$  has at least one more. So  $\text{rad}(b(n + 1))$  is a product of at least three distinct primes, none dividing  $a$ . By Proposition 3.8 and (7.1),

$$\text{rad}(b(n + 1)) \leq |J| < 2.1 a \ln a.$$

For  $a \leq 7$  the right side is less than  $29 < 30$ . For  $a = 8, 9, 10, 12$  the smallest possible values of  $\text{rad}(b(n + 1))$  are 105, 70, 231, 385, while  $2.1 a \ln a < 35, 42, 49, 63$  respectively.  $\square$

**Lemma 7.7.** *No factorization has  $a = 11$ .*

*Proof.* Suppose  $a = 11$ . As in Lemma 7.6,  $\text{rad}(b(n+1)) < 2.1 \cdot 11 \ln 11 < 56$ , and  $\text{rad}(b(n+1))$  is a product of at least three distinct primes different from 11. Hence  $\text{rad}(b(n+1)) \in \{30, 42\}$ .

If  $n$  is even, then  $n+1$  is odd and  $2 \mid b$ . Since  $n+1$  has at least two prime factors,  $b$  must be a power of 2 and  $3 \mid n+1$ . By Corollary 7.3,  $d$  is even, so Lemma 7.2(c) gives  $11^{n+1} \equiv 1 \pmod{3}$ . This is false, since  $n+1$  is odd.

So  $n$  and  $b$  are odd. Let  $q$  be the least prime factor of  $b$ . By Lemma 7.2(d), the order of 11 modulo  $q$  divides  $11b$ ; it also divides  $q-1$ , and  $\gcd(b, q-1) = 1$  by minimality of  $q$ . Hence the order divides 11. Order 11 would force  $q \geq 23$ , which is impossible. So  $q \mid 10$ , i.e.  $q = 5$ . Then  $\text{rad}(b(n+1)) = 30$ , and since  $n+1$  is not a prime power,  $b = 5^u$  and  $n+1 = 2^v 3^w$  with  $u, v, w \geq 1$ . Consider  $11 \cdot 5^u + 1 = 2^v 3^w$ :

- modulo 3,  $(-1)^{u+1} + 1 \equiv 0$ , so  $u$  is even;
- modulo 8,  $5^u \equiv 1$  gives  $v = 2$ ;
- modulo 5,  $4 \cdot 3^w \equiv 1$  gives  $w \equiv 2 \pmod{4}$ .

Therefore  $(2 \cdot 3^{w/2} - 1)(2 \cdot 3^{w/2} + 1) = 11 \cdot 5^u$ . The two factors on the left are coprime and greater than 1, so they are 11 and  $5^u$  in some order. They differ by 2, which would force  $5^u \in \{9, 13\}$ , a contradiction.  $\square$

**7.6. The range**  $182 \leq n < 1000$ . The lower bound of Lemma 5.3 used the full annulus of Lemma 2.1. Near  $n = 182$  that is too weak: the same argument would only give the constant  $\frac{1}{2} \ln L - \frac{3}{8}(L - \ln n) \approx 0.63$ , where  $L = \ln(2n+1)$ , below the  $\frac{33}{50}$  of Lemma 5.1. For  $n \geq 8$ , Lemmas 2.1 and 7.1(c) narrow the annulus: all the numbers  $\ln|\alpha|$  lie in an interval of length  $W/n$  with  $W = L - \frac{11}{6}$ , so (7.2) applies with this  $W$ .

**Lemma 7.8.** *Let  $d \leq (n-1)/2$ . If  $182 \leq n < 1000$  and  $a \geq 13$ , then*

$$\ln |\text{Res}(g, h)| > n \left( \frac{1}{8} \ln n + \frac{2}{3} \right) - \ln \frac{n(n+1)}{2}.$$

*Proof.* Write  $u = \ln a$  and  $L = \ln(2n+1)$ , so that  $u \geq \ln 13 > 2.55$  and, by (5.1),  $u < dL/n < L/2$ . By (7.2), it suffices to show  $G_W > \frac{1}{8} \ln n + \frac{2}{3}$ . We have  $5.2 < \ln n < 6.91$ ,  $0.69 < L - \ln n < 0.7$ , and  $1.4 < \ln W < 1.8$ . As a function of  $x = d/n$ ,  $G_W$  has derivative  $\ln n - u + \ln W - Wx > \ln W - (L - \ln n) > 0.7$ , using (5.1) and  $W < L$ . Hence

$$G_W > H_W(u) := \frac{u(\ln n + \ln W)}{L} - \frac{u^2(L + W/2)}{L^2}.$$

Moreover

$$H_W(u) - H_W(L/2) = \frac{u - L/2}{L} \left[ \ln n + \ln W - \left(u + \frac{L}{2}\right) \left(1 + \frac{W}{2L}\right) \right].$$

Using  $u \geq 2.55$ ,  $\ln W < 1.8$  and  $W = L - \frac{11}{6}$ , twice the bracket times  $L$  is at most  $L(2 \ln n - \frac{3}{2}L - \frac{47}{15}) + \frac{187}{40}$ . This decreases in  $L$ . At  $L = \ln n + 0.69$  it becomes  $(\ln n + 0.69)(\frac{1}{2} \ln n - \frac{2501}{600}) + \frac{187}{40}$ , which is increasing on  $[5.2, 7]$  and negative ( $< -0.46$ ) at 7. So the bracket is negative, and since  $u \leq L/2$  we get  $H_W(u) \geq H_W(L/2)$ . Finally,

$$H_W(L/2) = \frac{1}{8} \ln n + \frac{1}{2} \ln W - \frac{3}{8}(L - \ln n) + \frac{11}{48} > \frac{1}{8} \ln n + 0.7 - 0.2625 + \frac{11}{48} = \frac{1}{8} \ln n + \frac{2}{3}. \quad \square$$

## 7.7. Conclusion.

**Theorem 7.9.** *For  $2 \leq n < 1000$ , the polynomial  $f$  is irreducible; the proof below does not use the computation of Appendix A.*

*Proof.* Suppose a factorization exists. Then  $n \geq 6$ . Lemmas 7.6 and 7.7, applied to both labelings of the factors, give  $a, b \geq 13$ . As  $a$  and  $b$  are distinct and coprime,  $n = ab \geq 13 \cdot 14 = 182$ . Label the factors so that  $d \leq (n-1)/2$ ; both constant terms are still at least 13. Then Lemma 7.8 contradicts Lemma 5.1 (valid for  $n \geq 182$ ), since  $\frac{2}{3} > \frac{33}{50}$ .  $\square$

APPENDIX A. COMPUTATIONAL VERIFICATION FOR  $n < 1000$ 

For each  $2 \leq n < 1000$  we factored  $f$  over  $\mathbb{Z}$  with the FLINT library [1], through its Python interface `python-flint`, version 0.9.0. In every case the factorization consists of  $f$  itself, with multiplicity one, so  $f$  is irreducible. The whole computation takes about a minute and a half on a single core. The script is:

```
# Irreducibility over Q of f(x) = sum_{k=1}^n k x^{(n-k)}, 2 <= n < 1000.
# FLINT factors over Z; f is monic, so this decides irreducibility.
import flint
```

```
bad = []
for n in range(2, 1000):
    # coefficient of x^j in f is n - j
    f = flint.fmpz_poly([n - j for j in range(n)])
    assert f.degree() == n - 1 and f.coeffs()[-1] == 1
    content, factors = f.factor()
    (g, e), = factors if len(factors) == 1 else [(None, 0)]
    if not (e == 1 and g.degree() == n - 1):
        bad.append(n)
print("checked 2 <= n < 1000 ; reducible cases:", bad)
```

Its output is

```
checked 2 <= n < 1000 ; reducible cases: []
```

## REFERENCES

- [1] The FLINT team, *FLINT: Fast Library for Number Theory*, <https://flintlib.org>.
- [2] J. S. Milne, *Algebraic Number Theory*, course notes, version 3.01, 2008, <https://www.jmilne.org/math/CourseNotes/ANT301.pdf>.
- [3] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften 322, Springer, Berlin, 1999.